Photon Counting

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October 9, 2026

1 Introduction

How is the intensity of light determined? The intensity of a classical wave is determined by the amplitude of its oscillation and can, in principle, be reduced continuously. However, in the case of light, when the intensity is reduced, it eventually reaches a point where individual photons arrive sporadically. This behavior will be examined using a highly sensitive light sensor.

A photosensor produces a signal proportional to the light intensity. As the light becomes weaker, the signal from the sensor also becomes smaller. Beyond a certain point, however, the signal no longer decreases continuously: a signal either appears or does not appear. This is because light consists of photons, and the sensor cannot detect a fraction of a photon.

In this exercise, we will observe signals from the sensor using a small computer called Red Pitaya. We will study data processing and statistical analysis.

1.1 Setup

Please check the connections. The power to the MPPC module must be off (disconnected).

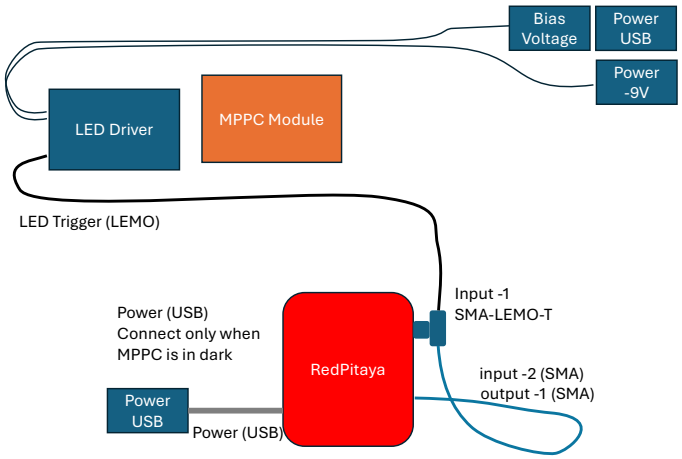


Figure 1: Setup for the measurements

2 RedPitaya

Red Pitaya is a small computer with analog inputs and outputs. Its name comes from the red pitaya, which is also known as the dragon fruit. In this exercise, a pulse output from the Red Pitaya triggers light emission from an LED driver. The analog signal input from the MPPC module will be measured.

The Red Pitaya boards are connected in the local network. Connect your PC to the Wi-Fi network "Buffalo-A-5430". The Wi-Fi password is provided in the room. Open a web browser (Google Chrome is recommended) and enter the IP address of the Red Pitaya. Please connect only one PC to each Red Pitaya. The IP address is labeled on each Red Pitaya.

MAC address	IP address
00:26:32:F0:9F:D5	192.168.11.21
00:26:32:F0:9F:E4	192.168.11.22
00:26:32:F0:9E:91	192.168.11.23
00:26:32:F0:9E:62	192.168.11.24
00:26:32:F0:74:86	192.168.11.25
00:26:32:F0:76:35	192.168.11.26

Direct Ethernet connection between the Red Pitaya board and your PC is also possible.

1. On Linux, open Network settings, go to Edit Connection and select Share to other computers under LAN network IPv4 settings.
2. Connect an Ethernet cable between your PC and the Red Pitaya board and wait approximately 2 minutes.
3. Open a web browser (Google Chrome is recommended) and enter rp-xxxxxx.local/ in the address bar. Here, xxxxxx represents the last six characters of the Red Pitaya board's MAC address.

A list of applications will appear in your web browser.



Figure 2: Red Pitaya main pages

We will use the "Photon Counting" application.
Select "Photon Counting". The application may appear on the second page rather than the first.

3 Light Pulser

The charge stored in a capacitor is discharged through an LED when triggered by an external signal, causing the LED to emit light.

This is similar to releasing water stored in a bucket (capacitor). Water is slowly stored (charged). If the water level (applied voltage) is high, more water (charge) is stored, and a larger current flows when it is discharged. The blue LED used in this exercise [1] requires a forward voltage of approximately 3.2 V or more for current to flow.

(Exercise) Calculate the energy of a blue photon (wavelength: 465 nm). To emit a blue photon, an electron must have at least that much energy. Compare this with the voltage applied to the light pulser.

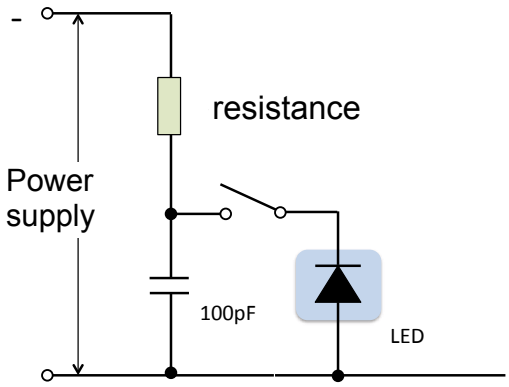


Figure 3: Operating principle of the LED driver

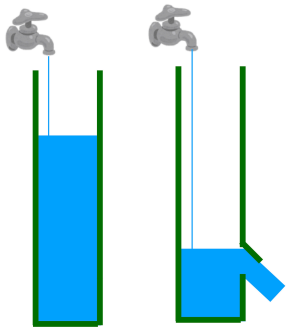


Figure 4: Analogy using a bucket of water

When the applied voltage exceeds 3.2 V, the excess voltage drives current through the LED, causing it to emit photons. For example, if 4.8 V is applied, the excess voltage is 1.6 V. The charge is given by $Q = CV$. With a capacitance of 100 pF, this corresponds to 160 pC of charge passing through the LED.

(Exercise) 160 pC of electrons flow through the LED. How many electrons is that?

You can go to the second page using the button at the bottom.

3.1 Observation with the Naked Eye

Connect a power supply (AC adapter) and a variable bias voltage supply to the LED pulser. We need a trigger signal for the LED from the Red Pitaya.

In the "Photon Counting" application, open the "MPPC Scope" page. Then select "Display: Coarse" and press the "Run" button. The count should increase. If it does

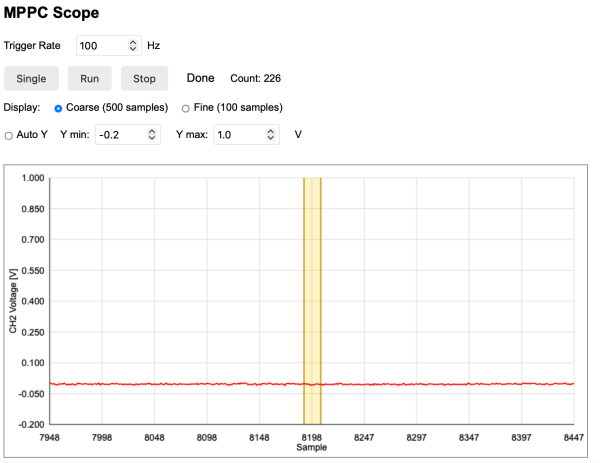


Figure 5: Red Pitaya / Photon Counting / MPPC scope page

not, please check the connections. The Red Pitaya is now generating the LED trigger signal. Set the LED trigger rate to 1000 Hz. In a dark room, you should see a blue glow at a bias voltage of approximately -11 to -13 V.

Find the minimum bias voltage at which you can see the blue light.

This light is too intense for the MPPC. Set the voltage to 7.0 V before proceeding to the next step.

3.2 Observation with MPPC (SiPM)

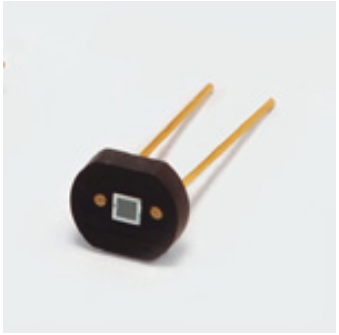


Figure 6: MPPC

4 MPPC (Multi-Pixel Photon Counter) / SiPM

An MPPC (SiPM) is a semiconductor sensor composed of many pixels. Each pixel is an avalanche photodiode. A single photon can trigger an electron avalanche. The device operates in Geiger mode, in which even a single photon can trigger a large output signal.

(Exercise) What is Geiger mode?

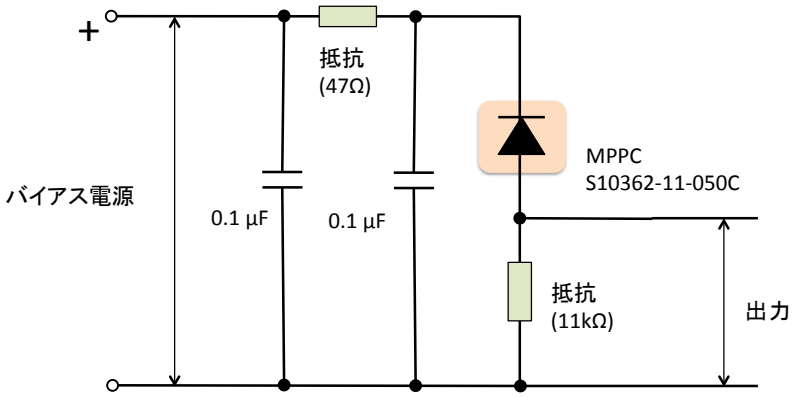


Figure 7: MPPC circuit diagram

The MPPC outputs the sum of signals from all pixels, allowing us to estimate how many photons were detected simultaneously. Since individual pixels are small, the probability of multiple photons hitting the same pixel is low under weak light, so the number of triggered pixels is proportional to the light intensity. The MPPC used in this exercise (S10362-11-050C) [2] has 400 pixels, and the output signal is proportional to the number of activated pixels. MPPCs are highly sensitive sensors—do not operate them under bright light.

4.1 Preparation

Like other diodes, MPPCs normally block current under reverse bias. Reverse bias means the anode has a lower potential than the cathode, preventing current flow. The MPPC used in this exercise requires approximately 70 V of reverse bias to operate. Although the dark current is small under reverse bias, photons can trigger avalanche pulses. The total current is proportional to the number of pixels that detected photons.

If the bias voltage is too low, the output signal is weak. If it is too high, the avalanche may not be quenched properly. To adjust this properly, use an MPPC module [3] with

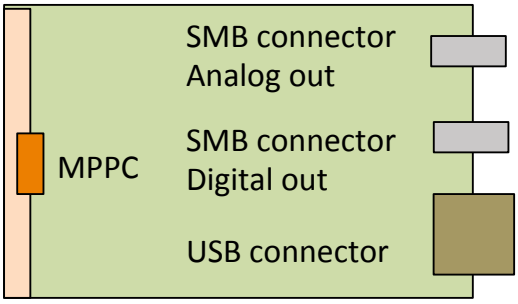


Figure 8: MPPC module

built-in voltage control. This module generates the bias voltage needed to maintain a stable output signal amplitude and is powered through USB (5 V).

Analog signals from this module will be used for the measurements below.

4.2 Cable Setup

Please connect the cables.

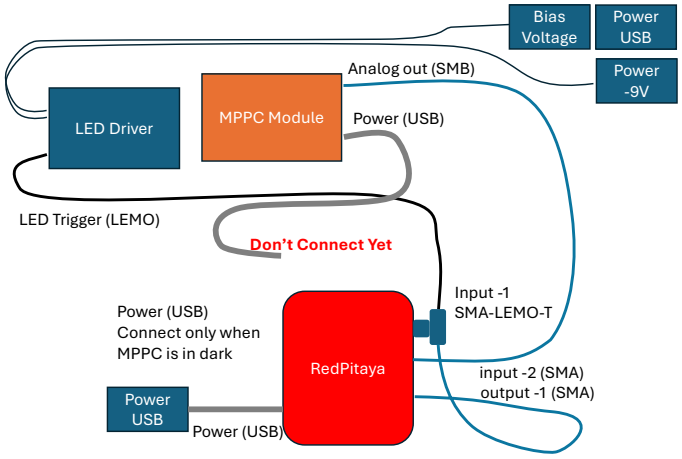


Figure 9: Setup for the measurements

Caution: Do not connect the MPPC power USB cable to the Red Pitaya at this stage.

This experiment deals with extremely weak light levels where individual photons can be counted. Measurements must be performed in a light-tight environment. Place the setup in a paper box and cover with a black sheet.

4.3 Module Operation

Before connecting the MPPC module, switch off the LED by disconnecting the -9 V supply to the LED driver. After confirming that the setup is properly shielded from light, connect the MPPC power USB cable to the Red Pitaya. Connect the analog signal to analog input 2 of the Red Pitaya. Connect the USB cable to the USB port of the Red Pitaya. This USB connection is used only to supply power.

4.4 Output Waveforms

In the "Photon Counting" application, open the "MPPC Scope" page. Then select "Display: Coarse" and press the "Run" button Observe the analog signal.

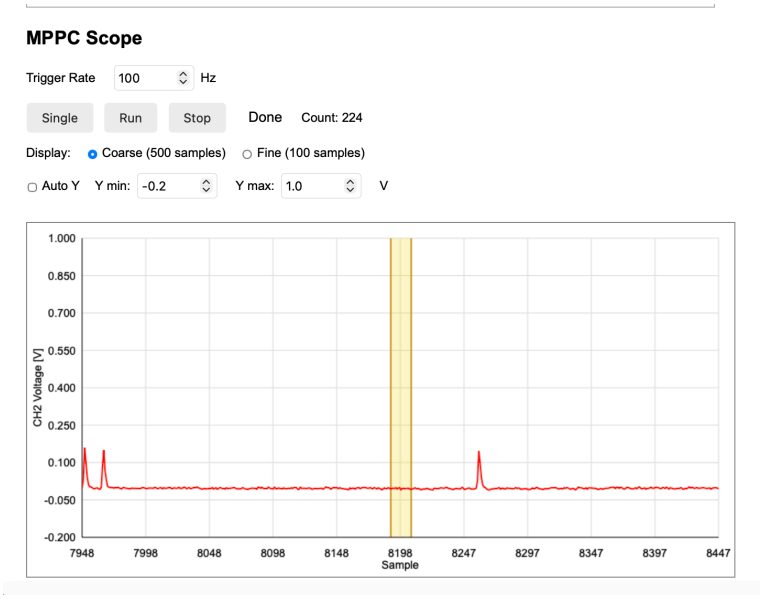


Figure 10: MPPC analogue output

You will see small signals (approximately 100 mV). These pulses are called "dark counts." MPPCs can have dark-count rates of approximately 100 kcps even when properly shielded from light. The coarse time range is 4 μs. We expect an average of 100 kHz $\times$ 4 μs = 0.4 counts. Thus, you will typically observe 0–2 pulses in this time range.

If more than 10 pulses are observed in 4 μs, light leakage may be the cause. You should improve light shielding.

4.5 MPPC Module Operation

To distinguish LED-induced signals from dark counts, we use a trigger. Trigger the oscilloscope using the signal from the function generator. Now connect the -9 V power supply to the LED driver. This makes photon-detection signals from the LED appear at consistent times.

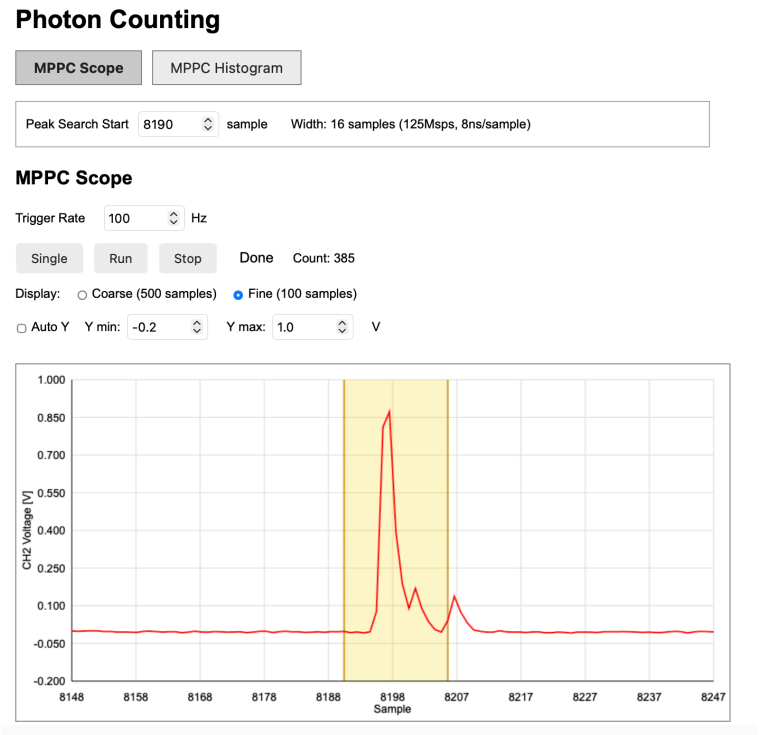


Figure 11: MPPC analogue output

Start with the LED voltage at approximately -6 V and gradually change it to -7 V. Dark counts occur randomly and rarely produce multiple simultaneous pulses. However, LED light produces many photons simultaneously, triggering multiple pixels at once. If you observe such multiphoton signals, note their timing (position on the screen). The input saturates at 1 V. If the LED intensity is increased too much, the signal will saturate.

Then gradually decrease the LED bias voltage. You will see the pulse height decrease, and eventually some triggers will produce no detected signal. This indicates that the average number of detected photons is so low that sometimes no photons are detected. Record the voltage at which this starts happening. As you lower the voltage further, signals will appear intermittently—record this voltage too.

5 Pulse Height Distribution

Next, we will measure the pulse-height distribution of the signals from the MPPC. We need to identify the signal position.

Make sure the signal is within the colored area. You can adjust the timing by changing the peak-position parameter.

Then go to the "Histogram" page. The "Run" button starts data accumulation.

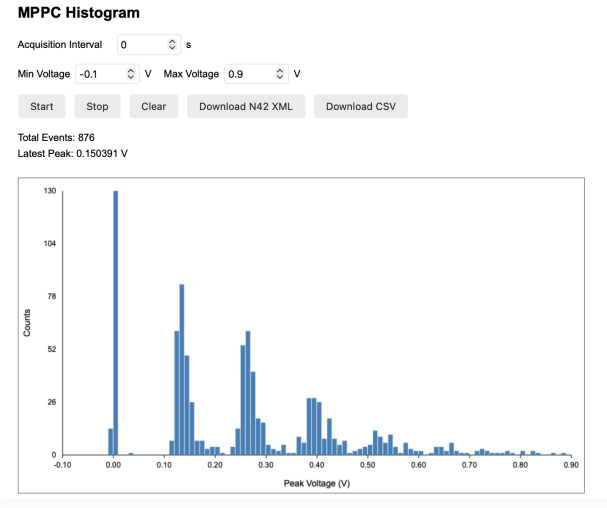


Figure 12: Pulse height distribution

In the histogram shown in the figure, the leftmost peak corresponds to no photons being detected, the next to one photon, and so on. From these peaks, we will determine the number of events for each detected photon count.

Note that each peak corresponding to a particular photon count has a finite width. One reason for this width is the relatively low sampling rate of the Red Pitaya.

5.1 Photon Number Distribution

After collecting a sufficient number of events, stop the acquisition. You can save the data as a CSV file or an N42 XML file. CSV files can be analyzed using Excel or other software. N42 files can be analyzed using "InterSpec". InterSpec is software developed at Sandia National Laboratories.

In Excel, you can sum the number of events for each photon count.

(Exercise) Obtain the average and variance of this distribution.

Suppose you obtain N_i events in which i photons are detected. The average number

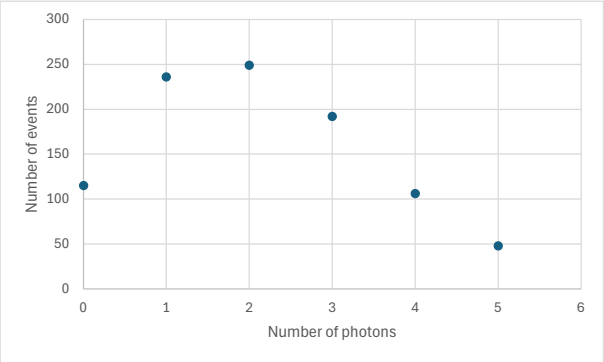


Figure 13: Photon number distribution

of detected photons is

$$\langle i \rangle = \frac{\sum i \cdot N_i}{\sum N_i} \tag{1}$$

Your calculated average may be smaller than the true average number of detected photons. This is because events with larger photon counts may not be included in this measurement.

The variance of this distribution is

$$\sigma^2 = \frac{\sum (i - \langle i \rangle)^2 \cdot N_i}{\sum N_i} = \frac{\sum i^2 \cdot N_i}{\sum N_i} - \langle i \rangle^2 \tag{2}$$

(Exercise) Compare the mean and the variance.

6 Poisson Distribution

The probability of a photon emitted from the LED being detected by the MPPC is very low. If N is the total number of emitted photons and p is the detection probability for each, then the probability P_n of detecting n photons follows the binomial distribution:

$$P_n = {}_N\mathrm{C}_n p^n (1-p)^{N-n} \tag{3}$$

Here, N is large and p is small, while the product $Np = \lambda$ remains finite (of order a few). We can rewrite:

$$P_n = \frac{N!}{n!(N-n)!} \left(\frac{\lambda}{N}\right)^n \left(1 - \frac{\lambda}{N}\right)^{N-n} = \frac{N!}{n!(N-n)!} \frac{\lambda^n (N-\lambda)^{N-n}}{N^N} \tag{4}$$

Taking the limit $N \to \infty$, the distribution depends only on λ:

$$\log P_n = \log N! - \log n! - \log(N-n)! + n\log\lambda + (N-n)\log(N-\lambda) - N\log N \tag{5}$$

Using Stirling ' s approximation for large factorials:

$$\log N! \approx N\log N - N \tag{6}$$

We simplify:

$$\begin{aligned} & \log P_n + \log n! - n\log\lambda \\ = \; & \log N! - \log(N-n)! + (N-n)\log(N-\lambda) - N\log N \\ = \; & -(N-n)\log(N-n) + (N-n)\log(N-\lambda) - n \\ = \; & (N-n)\log\left(\frac{N-\lambda}{N-n}\right) - n \\ = \; & (N-n)\log\left(1+\frac{n-\lambda}{N-n}\right) - n \end{aligned} \tag{7}$$

When x is small, $\log(1+x) \approx x$, so:

$$\log P_n \;\; = \;\; n\log\lambda - \log n! - \lambda \tag{8}$$

Therefore:

$$P_n = \frac{\lambda^n}{n!}e^{-\lambda} \tag{9}$$

This is known as the **Poisson distribution**.

6.1 Generating Function of the Poisson Distribution

Let ' s study properties of the Poisson distribution using the generating function. It is defined as:

$$G(x) = \sum_{n=0}^{\infty} P_n x^n \tag{10}$$

Substituting $x = 1$:

$$G(1) = \sum_{n=0}^{\infty} P_n = 1 \tag{11}$$

Differentiating with respect to x:

$$G'(x) = \sum_{n=1}^{\infty} nP_n x^{n-1} \tag{12}$$

$$G'(1) = \sum_{n=1}^{\infty} nP_n = \langle n \rangle \tag{13}$$

This gives the **mean**. Now, take the second derivative:

$$G''(x) = \sum_{n=2}^{\infty} n(n-1)P_n x^{n-2} \tag{14}$$

$$G''(1) = \sum_{n=2}^{\infty} n(n-1)P_n \tag{15}$$

The **variance** is:

$$\sigma^2 = \langle (n - \langle n \rangle)^2 \rangle = \langle n^2 \rangle - \langle n \rangle^2 = G''(1) + G'(1) - [G'(1)]^2 \tag{16}$$

Since:

$$G(x) = \sum_{n=0}^{\infty} P_n x^n = \sum_{n=0}^{\infty} \frac{(\lambda x)^n}{n!} e^{-\lambda} = e^{\lambda x - \lambda} \tag{17}$$

Then:

$$G'(x) = \lambda e^{\lambda x - \lambda}, \quad G'(1) = \lambda \tag{18}$$

$$G''(x) = \lambda^2 e^{\lambda x - \lambda}, \quad G''(1) = \lambda^2 \tag{19}$$

So:

$$\sigma^2 = \lambda^2 + \lambda - \lambda^2 = \lambda \tag{20}$$

Thus, for a Poisson distribution, the **mean equals the variance**.

7 Exercise

Verify that the measured photon number distribution follows a Poisson distribution. From the histogram of pulse heights, derive the photon number distribution. Check whether this distribution follows a Poisson distribution. Calculate the average photon number and express it as a function of the voltage applied to the LED. The MPPC size is 1mm × 1mm, and for 400 pixels, the detection efficiency is 35%. From this, estimate how many photons per 1mm^2 reach the MPPC.

8 Human Vision

MPPCs are highly sensitive light sensors. But how sensitive is the human eye as a light detector? Photoreceptor cells include cone cells (color-sensitive) and rod cells (high-sensitivity, brightness-sensitive). Rod cells are extremely sensitive and can respond to individual photons. Let ' s compare the sensitivity of the human eye with the MPPC.

How many photons per second must enter the eye for one to perceive light? The pupil size varies from 2 mm to 8 mm in diameter. In darkness, the pupil dilates. Let ' s assume a diameter of 8 mm, i.e., an area of about 50mm^2. This is 50 times the area of the MPPC. If the eye is placed seven times farther from the light source than the MPPC, the photon flux per unit area at the eye is 1/49 of that at the MPPC. Thus, the number of photons entering the eye is approximately the same as the number reaching the MPPC. With knowledge of the flash rate (per second), we can calculate the photon flux.

Note: This comparison is disadvantageous to the human eye because it is less sensitive to blue light. Human sensitivity to blue light is approximately one-tenth of that to green light.

References

1 LED Specification Sheet, Nichia Corporation, NSPB500S

2 S10362-11-050C
http://jp.hamamatsu.com/products/sensor-ssd/4010/4025/S10362-11-050C/index_ja.html
https://www.hamamatsu.com/resources/pdf/ssd/03_handbook.pdf

3 C10507-11-050C
http://jp.hamamatsu.com/products/sensor-ssd/4010/4026/C10507-11-050C/index_ja.html

4 http://www.home.agilent.com/agilent/product.jspx?nid=-34095.0.00&pageMode=OV&lc=eng&cc=VN#